

HSH HEAVY SAFE HELICOPTER

HSH-00 | REVISION 18

FINAL GENERATION 1 CONCEPTUAL / IP MASTER ENGINEERING BASELINE

Field	Value
Founder	Amirbahador Atai
Generation / Document	Generation 1 / HSH-00
Revision	Rev18
Maximum capsule occupancy	8 persons maximum
Vehicle target mass	26,000 kg
Safe capsule mass allocation	7,000 kg planning allocation
Main rotor target	26.0 m
Tail rotor target	4.0 m
Range target	1,000 km
Recovery study architecture	3 primary + 2 reserve canopies
Landing survival architecture	8 fixed energy-absorbing supports
Development stage	Pre-Simulation / Pre-Hardware / Pre-Prototype
Baseline status	FINAL G1 CONCEPTUAL MASTER BASELINE

Controlled status: This document is a conceptual and IP-development baseline. TARGET, ASSUMPTION, STUDY, CALC and OPEN values are not demonstrated performance. This document is not a certification approval, flight approval, production specification, or proof of achieved survival probability.

Revision 18 is the final Generation 1 conceptual master baseline. It unifies nomenclature to HSH-00 Rev18, locks maximum capsule occupancy at 8 persons, preserves the corrected safety architecture (no main-rotor separation in the core safety case), retains the staged separation timeline, the 3-primary + 2-reserve recovery study architecture, the eight fixed energy-absorbing landing supports, the Risk Register, the Prior Art Posture, the narrowed IP value surfaces, the Critical Open Items register, and the full engineering maturity framework. Commercial licensing packaging is intentionally kept outside this engineering baseline so that Document 00 remains a controlled technical master rather than a sales brochure. Future technical closure belongs in the controlled HSH Generation 1 engineering document tree.

1. REV18 CORRECTIVE BASELINE AND FREEZE

1.1 Controlled identity

The program name, document title and Generation 1 identity are unified as HSH Heavy Safe Helicopter. The controlled document is HSH-00, Revision 18, Founder: Amirbahador Atai. All prior internal references to HSC-00, Rev12, Rev13, Rev14, Rev15 or Rev16 as controlling identity are superseded by this Rev18 identity for document-control purposes. The engineering substance of the corrected architecture remains controlling.

1.2 Maximum capsule occupancy

The Generation 1 survival capsule has a hard configuration limit of 8 persons maximum. No higher-occupancy configuration belongs to this baseline. Any future change would require formal configuration control and re-analysis of geometry, mass, CG, inertia, seating, restraints, life support, egress, structural loads, recovery sizing and certification. The 8-person limit is a configuration constraint, not a claim that eight-person crash survival has already been demonstrated. That evidence remains a future engineering objective.

1.3 Baseline freeze

Rev18 is the final Generation 1 conceptual master baseline. Future engineering closure is to be performed in the numbered HSH Generation 1 document tree and traced back to this baseline. A future revision of HSH-00 should occur only for a material top-level architectural or configuration change.

1.4 Engineering maturity

G0 Architecture Freeze is the current baseline gate. G1 through G8 remain evidence gates for analytical closure, subsystem demonstration, integrated ground demonstration, dynamics validation, structural/crash testing, recovery testing, flight development and qualification/certification.

1.5 What Rev18 deliberately excludes from Document 00

Commercial licensing packaging (data-room scripts, term-sheet language, licensee negotiation phases, and marketing-oriented platform-family narrative) is not embedded in this engineering baseline. Those materials, if required, shall be controlled as a complementary commercial package referenced to this baseline, not as a substitute for engineering closure. This separation protects technical credibility under industrial due diligence.

2. TOP-LEVEL TECHNICAL BASELINE

Maximum Take-Off / Design Mass is 26,000 kg [TARGET]. Occupancy is 8 persons maximum [LOCKED LIMIT]. Flight architecture is a conventional single main rotor plus tail rotor [REQ/STUDY]. Main rotor diameter is 26.0 m [TARGET] with disk area 530.93 m² [CALC] and weight-based disk loading 48.97 kg/m² [CALC]. Tail rotor diameter is 4.0 m [TARGET] with disk area 12.57 m² [CALC]. Propulsion architecture is two turboshaft-class engines [REQ/STUDY]. Landing gear is not retractable wheeled gear; instead, eight short, thick fixed mechanical supports integrated with the capsule [TARGET/CONCEPT]. Range target is 1,000 km [TARGET]. Hover endurance is to be derived after power/fuel/thermal closure [OPEN]. The survival capsule is the dedicated occupied final landing boundary [REQ], with a preliminary complete mass of 7,000 kg [TARGET]. Capsule recovery uses a 3-primary / 2-reserve distributed canopy architecture [STUDY]. Rapid separation covers selective separation of designated high-mass/fire-hazard components excluding the main rotor [REQ/CONCEPT]. The separation development objective is expressed as a staged, per-subsystem timeline rather than a single figure [TARGET/OPEN per stage]. Main-rotor separation is not part of the core safety case [DEFERRED — RESEARCH TRACK].

Parameter	Rev18 Value	Status
Design mass	26,000 kg	[TARGET]
Main rotor diameter	26.0 m	[TARGET]
Main rotor disk area	530.93 m ²	[CALC]
Disk loading	48.97 kg/m ²	[CALC]
Tail rotor diameter	4.0 m	[TARGET]
Tail rotor disk area	12.57 m ²	[CALC]
Capsule mass	7,000 kg	[TARGET]
Range target	1,000 km	[TARGET]
Ideal hover power (study)	≈ 3.57 MW	[CALC]
Shaft power @ FM 0.75	≈ 4.76 MW	[CALC]

Parameter	Rev18 Value	Status
Occupancy	8 persons maximum	[LOCKED LIMIT]
Propulsion	Two turboshaft engines	[REQ / STUDY]
Main rotor separation	Not part of core safety case	[DEFERRED — RESEARCH TRACK]
Recovery canopies	3 primary + 2 reserve (5 total)	[STUDY]
Command-to-clear timeline	Staged per subsystem	[TARGET/OPEN per stage]

The 26.0 m rotor is intentionally sized for the 26-ton class; the resulting disk loading of approximately 48.97 kg/m² is a deliberate hover-margin and downwash-severity trade rather than an unexamined byproduct. Preliminary ideal hover power is approximately 3.57 MW at sea-level standard density; applying an illustrative figure-of-merit assumption of 0.75 gives approximately 4.76 MW rotor power — a study calculation only. Installed engine power must be determined from a complete power-required/power-available model including profile power, induced power, transmission losses, control margins, temperature, density altitude, rotor speed, transients and OEI requirements. The 1,000 km range is a design target; it shall not be presented as achieved range until the mission model closes with aircraft drag, rotor performance, engine specific fuel consumption, fuel capacity, reserves, payload, temperature/altitude and operational constraints.

3. MASS ACCOUNTING — 26,000 kg

The following planning mass groups close to the 26,000 kg design mass target. Values marked ASSUMPTION / STUDY require bottom-up closure with component data, uncertainty and growth tracking before any freeze claim.

Mass Group	Planning Mass	Status
Survival capsule, complete	7,000 kg	TARGET
Main airframe excluding capsule	5,000 kg	ASSUMPTION / STUDY
Main rotor, hub and controls	2,100 kg	ASSUMPTION / STUDY
Tail rotor, gearbox and drive	480 kg	ASSUMPTION / STUDY
Engines, transmission, gearboxes and mounts	3,100 kg	ASSUMPTION / STUDY
Fuel system + full usable fuel	3,400 kg	ASSUMPTION / STUDY
Rapid separation system	1,000 kg	ASSUMPTION / STUDY
Avionics, electrical, hydraulic, thermal, control	1,600 kg	ASSUMPTION / STUDY
Operating fluids and consumables	320 kg	ASSUMPTION / STUDY
Growth / configuration reserve	2,000 kg	PROGRAM RESERVE
TOTAL	26,000 kg	CALC / CLOSURE

3.1 Capsule component mass — 7,000 kg

The 7,000 kg capsule allocation is a planning target supported by the following bottom-up component estimate. It is not a demonstrated empty mass. Seat count is aligned to the locked 8-person maximum occupancy.

Capsule Component	Mass
Primary crashworthy cell / internal roll-cage	1,950 kg
Outer capsule shell / fairing / thermal protection	500 kg
Eight fixed landing supports + energy absorbers	850 kg
Crashworthy seats + restraints, 8 positions	700 kg
Emergency exits / doors, frames and seals	250 kg
Distributed multi-canopy recovery system	650 kg
Independent emergency power + batteries	250 kg
Life support / oxygen / environmental	200 kg
Avionics / sensing / communications / locator	250 kg
Separation interfaces + release hardware	350 kg
Internal secondary structure / floor / padding	500 kg
Fire / smoke detection and suppression	150 kg
Water landing / flotation / sealing	150 kg

Capsule Component	Mass
Inspection / access / replaceable protection	100 kg
Capsule configuration growth allowance	150 kg
TOTAL	7,000 kg

3.2 Detailed aircraft mass breakdown (retained planning detail)

Main Airframe — 5,000 kg: Fuselage primary load-bearing structure 1,450 kg; frames, bulkheads and longerons 650 kg; tail boom and tail structural assemblies 520 kg; engine and transmission support structure 380 kg; rotor mast/transmission support integration structure 260 kg; external skins, fairings and aerodynamic panels 310 kg; doors, access panels and hinges 190 kg; firewalls, barriers and structural fire protection 170 kg; structural fittings, lugs, fasteners and local reinforcements 260 kg; cabin/service floor structure outside capsule boundary 210 kg.

Main Rotor System — 2,100 kg: Main rotor blades (26.0 m system) 980 kg; rotor hub and blade retention/grips 280 kg; mast and rotor-head structural components 260 kg; swashplate and pitch-control hardware 170 kg; pitch links, control rods and bearings 110 kg; rotor vibration/dynamic-control hardware 80 kg; rotor lubrication/inspection/access hardware 40 kg; configuration allowance 40 kg.

Tail Rotor System — 480 kg: Tail rotor blades (4.0 m system) 145 kg; tail rotor hub and blade retention 65 kg; tail rotor gearbox 95 kg; drive shafting, couplings and bearings 60 kg; remaining allocation to mounts, fairings and allowance.

4. PURPOSE AND SCOPE

HSH Generation 1 defines the system-level architecture for a 26-ton-class single-main-rotor helicopter whose primary distinguishing engineering concept is an occupied survival capsule that can become the final landing boundary during a catastrophic emergency. The aircraft is designed around a layered survival chain rather than reliance on a single emergency device. The survival chain is designed to work correctly under the assumption that the main rotor is never separated from the aircraft. This is a deliberate scoping decision made because no credible trajectory, gyroscopic, or structural analysis yet exists to support main-rotor separation, and building the core safety case on an unproven capability would undermine confidence in the entire architecture.

The baseline covers the aircraft, occupied survival capsule, rapid selective separation system (non-rotor components), safety initiation paths, deterministic safety authorization, stabilization, recovery systems, impact attenuation, post-event survival and the interfaces required to verify the complete chain. Separation is a controlled emergency mode authorized only when defined conditions are satisfied. The survival architecture is intended to support multiple emergency outcomes: conventional flight recovery where possible, autorotation where applicable, selective hazard/mass reduction, capsule separation, parachute-assisted descent, energy-absorbing landing and post-landing rescue. The architecture must not require the most extreme survival mode for every failure.

The baseline also preserves human emergency authority. Automatic systems may initiate or recommend a survival sequence only within defined authorization envelopes; the crew must retain an independent means to command, inhibit or terminate automatic action where technically and physically possible. This document is an engineering/IP baseline, not a certification approval. Airworthiness, structural substantiation, parachute qualification, release reliability, occupant injury criteria, flight performance and global patent novelty remain open work items.

5. BASELINE STATUS AND CONFIGURATION CONTROL

Status Label	Meaning in this document
[LOCKED CONCEPT]	Architecture intentionally retained as a baseline design principle; implementation remains subject to engineering verification
[REQ]	Top-level requirement proposed for subsequent requirement allocation
[TARGET]	Numerical or functional development objective; not demonstrated
[ASSUMPTION / STUDY]	Working engineering assumption used for preliminary sizing/accounting
[CALC]	Value calculated from stated assumptions and equations
[OPEN]	Requires analysis, simulation, test, supplier data or design decision
[OPEN — CRITICAL]	An open item with material effect on the core safety case; flagged for priority closure
[DEFERRED — RESEARCH TRACK]	Explicitly excluded from the licensable core baseline pending independent research
[TBD — DEPENDENCY]	Cannot be responsibly fixed until a dependent design activity closes
[LOCKED LIMIT]	Hard configuration constraint for Generation 1 (e.g., 8-person occupancy)

The baseline shall be configuration-controlled. Any change to MTOW, rotor diameter, capsule mass, occupant count, separation sequence, parachute architecture, landing-support geometry or safety-layer independence shall trigger an

impact assessment across mass properties, CG, inertia, dynamics, structures, controls, recovery and verification.

6. AIRCRAFT GEOMETRY AND CONFIGURATION

The geometry is a preliminary architecture sized around the 26-ton class. Dimensions other than rotor diameters are deliberately kept as ranges or open items until aerodynamic, structural, cabin, capsule and systems layouts are reconciled. The 26.0 m main rotor and 4.0 m tail rotor are locked targets for Generation 1 planning. Capsule external geometry remains subject to the 8-person occupancy lock, energy-absorbing support layout, exit arrangement and recovery-system packaging.

7. SIX-LAYER SURVIVAL AND SAFETY ARCHITECTURE

The defining engineering proposition remains the integration of multiple survival mechanisms into one state-aware architecture. The aircraft is not intended to depend solely on a parachute, solely on autorotation, solely on a detachable cabin or solely on automatic software. The intended chain is:

PREVENT/MONITOR → DETECT → CONFIRM → ASSESS → RECOVER IF POSSIBLE → SAFETY AUTHORIZE → SELECTIVELY SEPARATE (NON-ROTOR) → VERIFY CAPSULE CLEARANCE → DEPLOY RECOVERY → MANAGE DESCENT → ABSORB IMPACT → SURVIVE/LOCATE/RESCUE → AUDIT.

Layer independence, common-cause control and degraded-mode behaviour are first-class design concerns. No single sensor, power bus, software partition or structural attachment shall be assumed sufficient alone to complete the survival chain.

8. DETERMINISTIC SAFETY AUTHORIZATION AND SAFETY KERNEL

The Safety Kernel is the authorization boundary between complex detection/decision logic and physical survival actuation. AI, predictive models and optimizers may assist monitoring, detection and candidate-action ranking, but cannot bypass deterministic safety preconditions. Kernel inputs should include altitude, airspeed, attitude and rates, rotor state, engine/fire/fuel state, structural health, capsule attachment health, separation clearance, recovery-system availability, sensor validity, power availability and human authority state.

Kernel decision	Minimum design question	Verification
Authorize release	Are all mandatory preconditions true?	Formal/structural logic review + test
Inhibit release	Can an unsafe state block actuation?	Fault injection
Fallback	What happens if a sensor/path is lost?	Failure testing
Human override	Is authority bounded and deterministic?	HMI + system test
Audit	Can the event be reconstructed?	Data-recording test

9. CONTROLLED RAPID SEPARATION SYSTEM

The separation architecture is selective and staged. Candidate non-rotor components include engines, selected transmission/drive components, tail rotor/drive components and isolated fuel hazards. Main-rotor separation is explicitly excluded from the core safety case and retained only as a labeled, walled-off future research track. Each release event requires positive retention in normal flight, electrical/logic inhibition, independent command paths, release-state sensing, energy isolation, asymmetric-release detection, clearance verification and a defined fallback.

9.1 Staged separation timeline

A single unsupported “≤6 s from validated command to verified capsule-clear state” target is not used. Rev18 retains the staged, per-subsystem timeline that shows its work and flags its own weakest link:

Stage	Action	Target window	Confidence basis
S0	Validate emergency state; arm authorized path; isolate fuel/electrical/energy hazards	0.0–0.5 s	Comparable to modern ejection-seat arming logic
S1	Fuel isolation and designated fire-hazard separation	0.5–1.8 s	Comparable to certified crash fuel-shutoff systems
S2	Engine and tail-rotor drive separation (non-main-rotor)	1.8–3.5 s	Comparable to external cargo-release hardware timing, scaled
S3	Capsule-to-airframe release	3.5–5.0 s	[OPEN — CRITICAL] — no direct engineering analog; flagged honestly
S4	Capsule clearance verification	5.0–5.8 s	Sensor-response-limited, not mechanism-limited

Stage	Action	Target window	Confidence basis
S5	Recovery canopy authorization	After S4, gated on verified clearance	Not time-gated; correctness prioritized over speed

If detailed architecture cannot safely achieve these windows, targets shall be revised under configuration control rather than achieved by reducing verification rigor. Partial and asymmetric separation remain first-class emergency states at every stage — the system shall not assume that every release mechanism functions perfectly.

9.2 Core safety case — rebuilt without main-rotor separation

The survival chain is designed to work assuming the main rotor is never separated. Revised primary survival chain:

1. Controlled flight recovery / autorotation.
2. Selective separation of non-rotor hazards only: engines, fuel tankage (post-isolation), tail rotor/drive assembly, designated fire-load panels.
3. Capsule release from the airframe — the capsule departs from beneath/behind the still-attached (or autorotating) main rotor disk, not through it.
4. Capsule-only descent under the distributed recovery canopy system.
5. Impact attenuation via the eight fixed supports.
6. Post-landing survival and locate/rescue.

Main-rotor separation is retained as a labeled future research item, explicitly excluded from the licensable core baseline and from the primary IP claims. This prevents a buyer's diligence team from discounting the entire package on the strength of its single weakest claim, while preserving the option to reintroduce a rotor-separation capability later if independent 6-DOF, structural and trajectory analysis demonstrates it is achievable.

10. OCCUPIED SURVIVAL CAPSULE

The capsule is the dedicated occupied final landing boundary, with a continuous CFRP roll-cage concept, metallic/titanium nodes, energy-absorbing seats for 8 positions, eight fixed supports and independent emergency functions after separation. The capsule program shall establish load paths from ground contact through energy absorbers, lower structure, primary cell, floor, seat attachments and restraints. Structural cases shall include vertical, longitudinal, lateral and combined loading, asymmetric touchdown, failed supports, slope and uneven terrain.

Subsystem	Primary question	Evidence required
Roll-cage	Can the cell maintain a survivable occupied volume?	FEA + structural test
Joints/nodes	Can interfaces transfer combined crash loads?	Coupon/component test
Floor	Can loads reach the intended absorbers without local failure?	FEA + test
Seats/restraints	Can occupant loads be managed within criteria?	Dynamic test
Attachment fittings	Can normal and emergency load paths be controlled?	Static/fatigue/test

11. EIGHT FIXED SHORT MECHANICAL LANDING SUPPORTS

The eight supports are part of the survival system rather than conventional retractable landing gear. Their redundancy is an engineering hypothesis requiring asymmetric failure analysis, terrain interaction analysis and physical drop testing. The energy absorber shall have a defined force-displacement response, inspection criteria and replacement philosophy. The analysis set shall include all combinations that can materially change capsule attitude, peak loads or tipping resistance.

Case	Analysis focus	Required evidence
All supports available	Load sharing and stability	FEA + drop test
One support degraded	Redistribution and local overload	FEA + test
Multiple support degradation	Survivability boundary	Sensitivity study
Slope	Roll/tip tendency and load transfer	Terrain simulation
Soft ground	Penetration and energy dissipation	Ground interaction test

12. RECOVERY ARCHITECTURE — 3 PRIMARY + 2 RESERVE CANOPIES

An undifferentiated eight-canopy concept is not used. Rev18 retains a structure closer to flight-proven capsule-recovery precedent: three larger primary canopies sized to the 7,000 kg capsule using standard aerospace descent-rate targets, plus two independent reserve canopies deployed only on detected primary-canopy failure. Fewer, larger canopies reduce inflation-sequencing complexity and inter-canopy collision risk at low altitude — the failure mode most likely in real rotorcraft

emergencies. No canopy count shall be treated as proof of recovery performance. The system must be analyzed as a coupled capsule, harness, riser, canopy and atmosphere problem, including deployment transients and asymmetric failures. The minimum deployment altitude at which this (or any) capsule recovery system can be relied upon remains [OPEN — CRITICAL].

Recovery mode	Use case	Status
Controlled flight recovery	Recoverable control/propulsion disturbance	[REQ/STUDY]
Autorotation	Engine/power loss when altitude/rotor state permit	[REQ/STUDY]
Selective mass/hazard separation	Fire, structural or combined emergency requiring hazard/mass reduction	[REQ/CONCEPT]
Capsule separation	Catastrophic state where independent occupied recovery is preferable	[REQ/CONCEPT]
3 primary recovery canopies	Primary post-separation vertical recovery	[STUDY]
2 reserve canopies	Deployed only on detected primary failure	[STUDY]
Eight energy-absorbing supports	Final ground impact attenuation	[REQ/TARGET]
Post-landing survival	Locator, communications, lighting, emergency power, rescue interface	[REQ]

13. EMERGENCY STATE MACHINE

Explicit states include Normal Flight, Anomaly Detection, Confirmation, Impact Assessment, Recovery Attempt, Survival Authorization, Staged Separation, Capsule Clearance, Recovery Deployment, Descent Management, Landing Attenuation, Post-Landing Survival and Audit. Every transition should identify entry conditions, exit conditions, inhibited transitions, sensor validity requirements, crew authority, time constraints and degraded behavior.

State	Entry condition	Exit / next state
Normal	Nominal monitoring	Anomaly detected or normal continuation
Confirmation	Potential abnormality	Confirmed or rejected condition
Recovery attempt	Safe conventional recovery available	Recovery success or survival authorization
Survival authorization	Defined catastrophic conditions	Staged separation or inhibited state
Capsule clearance	Release executed	Recovery deployment authorization
Post-landing	Ground contact / stabilized	Rescue and audit complete

14. INDEPENDENCE, FAILURE MODES AND COMMON-CAUSE CONTROL

Identified seed failure modes include unintended separation, partial separation, asymmetric separation, release failure, sensor loss, automatic detection failure, power loss, communication loss, AI/model failure, fire during separation, released-component collision, parachute deployment outside envelope and landing-support failure. These form the seed set for FMEA/FMECA, fault-tree analysis, common-cause analysis, zonal safety analysis and particular-risk analysis. Independence must be demonstrated against common power, wiring, software, sensors, structure, fire zones and environmental exposure.

Failure mode	Immediate effect	Design response
Unintended release	Loss of structural attachment	Mechanical locks + inhibits + monitoring
Partial release	Asymmetric mass/inertia	State sensing + degraded recovery
Sensor disagreement	Unsafe authorization risk	Validity logic + diverse sensing
Power loss	Loss of survival functions	Independent emergency power
AI/model failure	Bad recommendation	Kernel remains authoritative
Fire	Loss of structure or fuel hazard	Isolation + containment + sequencing

15. PRINCIPAL SYSTEM INTERFACES

Interface	Definition	Control requirement
CAP-AF	Capsule ↔ airframe structural attachment	Normal-flight load path + controlled emergency release + positive retention
CAP-PAR	Capsule ↔ parachute system	Deployment only after validated capsule state and clearance

Interface	Definition	Control requirement
CAP-LG	Capsule ↔ eight landing supports	Primary crash/landing load path with energy absorption
JET-PWR	Independent release power	Segregated from common single-point power failure where required
JET-CMD	Automatic/backup/manual commands	Independent paths and protected actuation
JET-SNS	Release-state sensing	Positive confirmation of attached/released state
FUEL-ISO	Fuel isolation ↔ separation	Isolation before or coordinated with release per hazard analysis
REC-LOC	Post-landing locator/communications	Independent emergency power and antenna strategy
AV-CTRL	Avionics/control ↔ Safety Kernel	Deterministic authorization boundary

All interfaces crossing survival boundaries shall be documented in Interface Control Documents and assessed for common-cause vulnerability. Connector, harness, hydraulic, fuel and structural interfaces shall be designed so that emergency separation does not create an uncontrolled hazard.

16. VERIFICATION, VALIDATION AND EVIDENCE PROGRAM

Evidence gates are required before any high-confidence safety or performance claim is made. As of this revision, no gate beyond G0 (Architecture Freeze) is complete. This is stated plainly.

Gate	Evidence objective	Exit condition
G0	Architecture Freeze — safety philosophy, system boundaries, interfaces, emergency states	Baseline frozen
G1	Analytical Closure — mass, CG, inertia, rotor, power, fuel, structural paths, separation kinematics	Analytical closure acceptable
G2	Subsystem Demonstration — release mechanisms, locks, absorbers, parachute components, sensors, power	Critical functions demonstrated
G3	Integrated Ground Demonstration — end-to-end sequence without flight exposure	End-to-end ground evidence
G4	Dynamic / 6-DOF Simulation — vehicle and capsule dynamics, partial/asymmetric release, inflation	Model correlation and uncertainty bounded
G5	Structural / Crash Testing — capsule, attachments, seats, supports, absorbers, occupant protection	Defined structural cases closed
G6	Recovery System Testing — canopy deployment, inflation loads, failure cases, stability, landing	Recovery envelope demonstrated
G7	Flight Development — controlled flight tests within approved envelope	Flight envelope progressively established
G8	Certification / Qualification — evidence package against selected certification basis	Certification/qualification basis satisfied

Terms shall be used carefully: “specified” means defined in the design; “analyzed” means a documented analytical result; “simulated” means supported by a model; “tested” means demonstrated in a defined test; “verified” means evidence shows compliance with a requirement; “validated” means suitability has been established for the intended use; “qualified” and “certified” require the applicable formal processes.

17. ENGINEERING / IP VALUE SURFACES

Nine equally weighted IP assertions dilute attention and invite a diligence team to discount the whole set once the weakest are found unpatentable or over-broad. Rev18 concentrates on three most defensible claim candidates and retains the rest as supporting system narrative.

17.1 Primary claim candidates

IP-A — Deterministic Safety Kernel gating AI-assisted emergency prediction from physical actuation. The combination of (a) AI/ML-based anomaly prediction, (b) a hard deterministic precondition layer, and (c) selective, state-dependent (not all-or-nothing) actuation is a specific systems-engineering combination, not a single known device. This is assessed as the strongest claim in the package.

IP-B — Eight-point fixed capsule-integrated landing support with independent asymmetric/partial-failure state management. Combining the structural configuration (eight-point, capsule-integrated, non-retractable, energy-absorbing) with explicit failure-state detection and degraded-recovery logic as a single claim is narrower and more defensible than treating the hardware and the logic as separate elements.

IP-C — State-dependent, non-rotor selective separation sequencing for an occupied survival capsule. By explicitly excluding main-rotor separation, this claim is narrowed to something closer to precedented selective-jettison systems with a specific new element: capsule-centric, staged, state-dependent sequencing logic rather than an unbounded claim over “separate anything.”

The six-layer survival taxonomy, the multi-mode recovery chain, state-aware partial/asymmetric-release management, the optional residual-airframe recovery concept, and the immutable emergency event record are retained as descriptive system architecture. They strengthen understanding of the overall product but are not presented as independently patentable. No novelty, patentability or freedom-to-operate conclusion is asserted by this baseline.

18. RISK REGISTER

The risk register is not a declaration of safety. It is a mechanism for identifying where the concept can fail and where evidence must be generated before a safety claim is made. A buyer’s engineering diligence team will identify these risks regardless of whether they are disclosed. Naming them here is what makes the surrounding technical claims credible rather than promotional.

Risk	Severity	Current status	Priority
Capsule-to-airframe release (Stage S3) has no direct engineering precedent	High	Open research item, explicitly flagged	Critical
Low-altitude canopy deployment envelope may exclude many real accident profiles	High	Open — needs minimum-altitude closure study	Critical
7,000 kg capsule mass not yet bottom-up verified	Medium	Planning allocation only	Critical
No 6-DOF simulation evidence exists yet for any separation stage	High	Gate G4 not started	Critical
No professional prior-art search has been conducted	Medium	Prior-art posture only	High
Certification basis (CS-29 analog) not formally engaged with any regulator	Medium	Conceptual reference only	High
Main-rotor separation capability, if later reintroduced, remains structurally unproven	Medium	Deferred to research track	Medium

19. PRIOR ART POSTURE

This section does not assert novelty conclusions — that requires a professional prior-art search — but names the nearest known analog classes so a reviewer sees baseline awareness rather than legal naivety:

Ejection-seat-equipped rotorcraft. Existing rotorcraft designs (e.g., certain Soviet/Russian attack helicopters) use rotor-blade-severing explosive charges paired with a crew ejection seat. This is directly relevant prior art for any main-rotor-related separation claim, which is precisely why main-rotor separation is removed from the core claim set.

Crewed spacecraft return capsules. Multi-canopy recovery, crashworthy seating and independent post-landing life support in existing crewed spacecraft are relevant prior art for the capsule-recovery and life-support elements. IP language is narrowed to the rotorcraft-specific combination rather than the general capsule-recovery concept, which is not novel on its own.

Existing crash-attenuating landing gear / energy-absorbing strut systems in certified rotorcraft are relevant to the eight-support claim (IP-B). Novelty must rest on the capsule-integration and failure-state-management aspects specifically, not on energy-absorbing landing gear in general. A professional prior-art search against IP-A, IP-B and IP-C specifically remains a prerequisite before any patent filing or broad data-room disclosure.

20. CRITICAL OPEN ITEMS REGISTER

The following open items materially affect the credibility of the Generation 1 baseline. They are listed explicitly so that architecture, risk and verification remain aligned. These items are not weaknesses to be hidden; they are the work packages that must be closed before high-confidence safety or performance claims are made. Status labels follow Section 5.

ID	Open item	Status	Priority	Closure path
OI-01	Capsule-to-airframe release mechanism (Stage S3) has no direct certified engineering analog	OPEN — CRITICAL	P0	Mechanism design, 6-DOF, ground demo
OI-02	Minimum reliable canopy deployment altitude for 7,000 kg capsule recovery not closed	OPEN — CRITICAL	P0	Sizing + inflation dynamics + envelope study
OI-03	7,000 kg capsule mass remains a planning allocation, not a bottom-up verified empty mass	OPEN — CRITICAL	P0	Component mass database + weighing plan

ID	Open item	Status	Priority	Closure path
OI-04	No 6-DOF simulation evidence exists yet for any separation stage	OPEN — CRITICAL	P0	Model hierarchy + correlation plan (G4)
OI-05	Partial and asymmetric separation detection and degraded recovery logic not yet detailed	OPEN — CRITICAL	P0	State sensing + FMEA + test cases
OI-06	Independence of the six safety layers against common-cause failures not yet demonstrated	OPEN — CRITICAL	P0	CCA + fault injection + power/sensor diversity
OI-07	Occupant injury criteria and seat/restraint dynamic performance for 8 positions not closed	OPEN	P0	Dynamic seat tests + criteria selection
OI-08	Eight-support asymmetric failure and terrain interaction cases not closed	OPEN	P0	FEA + drop test matrix
OI-09	26,000 kg vehicle mass model lacks component-level uncertainty and growth tracking	OPEN	P1	Mass control plan + reserve policy
OI-10	Rotor/propulsion power-available vs power-required model (incl. OEI) not closed	OPEN	P1	Performance model + engine data
OI-11	Fuel isolation timing coordinated with separation sequence not closed	OPEN	P1	Hazard analysis + sequence timing
OI-12	Professional prior-art search against IP-A, IP-B and IP-C not yet performed	OPEN	P1	External IP counsel search
OI-13	Certification basis (CS-29 analog or equivalent) not formally engaged	OPEN	P1	Regulator engagement plan
OI-14	Interface Control Documents for CAP-AF, CAP-PAR, CAP-LG and JET-* not yet issued	OPEN	P1	ICD set under configuration control
OI-15	Main-rotor separation remains deferred research; no analysis to reintroduce it exists	DEFERRED — RESEARCH TRACK	P2	Only if independent research justifies re-entry

No item in this register is closed by the existence of this Document 00. Closure requires analysis, simulation, test evidence or formal design decision under configuration control, as applicable. P0 items should be addressed before any claim of integrated survival performance is presented outside a controlled technical evaluation environment.

21. ENGINEERING BASIS AND EXTERNAL REFERENCE NOTES

Current EASA CS-29 material was reviewed as an external engineering reference. The EASA material emphasizes that jettisonable configurations require satisfactory separation characteristics, including clear trajectory, no damaging interaction, no adverse/uncontrollable aircraft reaction, acceptable post-jettison stability/control and acceptable post-jettison performance; the verification program therefore treats these as core analysis/test concerns rather than assuming safe release. External reference: EASA, Easy Access Rules for Large Rotorcraft (CS-29).

22. DESIGN PHILOSOPHY — CONTROLLED SURVIVAL, NOT A SINGLE DEVICE

The defining engineering proposition of HSH Generation 1 remains the integration of multiple survival mechanisms into one state-aware architecture. A safety case that quietly needs its riskiest, least-proven capability to succeed is not actually a layered safety case. Removing main-rotor separation from the core safety case strengthens, rather than weakens, the layered proposition. The strongest future evidence will come from a closed mass model, rotor/propulsion analysis, 6-DOF simulation, deterministic safety logic, separation mechanism tests, capsule structural tests, parachute tests and progressively integrated demonstrations.

23. ENGINEERING MATURITY FRAMEWORK

23.1 Configuration and design principles

Configuration item	Rev18 control rule	Closure evidence
MTOW / design mass	Freeze only after bottom-up mass model and reserve policy are reconciled	Mass report + uncertainty analysis
Rotor geometry	Maintain 26.0 m target until performance study closes	Rotor sizing + power model
Capsule mass	7,000 kg planning target; no closure claim	Bottom-up mass budget
Occupancy	8 persons maximum locked for Generation 1	Configuration control

Configuration item	Rev18 control rule	Closure evidence
Recovery	3 primary + 2 reserve study architecture	Sizing + inflation dynamics + failure testing
Separation	Non-rotor candidate set only in core baseline	6-DOF + clearance + ground demonstration
Safety authorization	Deterministic kernel remains authoritative	Requirements + software/hardware verification

23.2 Mass, CG and inertia closure

The 26,000 kg planning mass and 7,000 kg capsule allocation are configuration-controlled hypotheses requiring bottom-up closure. The mass model should be maintained at component level, with uncertainty and growth tracked separately from nominal mass. The same dataset should feed center-of-gravity and inertia calculations for normal flight, emergency configurations, capsule separation and partial/asymmetric separation. A separation command cannot be evaluated independently of the resulting mass properties.

23.3 Main rotor and flight performance development

Performance closure should include induced power, profile power, parasite power, transmission losses, engine power available, density altitude, temperature, rotor RPM, transient margins, OEI operation and autorotation behavior. The rotor disk loading trade should remain open until the dedicated performance study closes. Hover-power figures in this document remain study values only.

23.4 Propulsion, transmission and energy

The baseline specifies two turboshaft-class engines as a proposed architecture. Explicit separation is required between engine selection, installation, transmission sizing, fuel system, thermal limits and emergency isolation. Fuel isolation is an enabling safety function. Isolation, venting, fire detection, engine shutdown, release sequencing and thermal management must be coordinated without assuming that any single subsystem will always respond correctly.

23.5 6-DOF dynamics and separation model

The model hierarchy shall represent mass properties, aerodynamic forces, rotor effects, control authority, release impulses, component trajectories, capsule attitude/rates and recovery-system deployment. Uncertainty should be propagated rather than hidden. Required model blocks include aircraft pre-release dynamics, capsule release 6-DOF trajectory and clearance, released-component collision envelope, partial-release CG/inertia changes, parachute inflation transients and ground-contact kinematics.

23.6 Master verification matrix fields

Requirement evidence field	Required content
Requirement ID	Unique controlled identifier
Source	Parent requirement / architecture reference
Allocation	System / subsystem owner
Verification method	Analysis, inspection, demonstration, test
Configuration	Applicable hardware/software revision
Result	Pass, fail, conditional, open
Evidence	Report, dataset, test record or approved analysis
Discrepancy	Open NCR/deviation if applicable
Approval	Responsible engineering authority

23.7 Certification and compliance readiness

This document is not certification approval. Conceptual architecture is separated from certification evidence. The correct objective at this stage is traceability: every future certification claim should have an identified rule or certification objective, means of compliance, verification method, responsible owner and evidence record.

23.8 Human factors and crew authority

Human emergency authority is preserved. The crew should understand system state, inhibited actions, available recovery modes and the consequences of manual intervention. Emergency controls should be physically and logically separated from routine controls where justified. Human authority must remain bounded by physical safety interlocks so that manual action cannot silently defeat mandatory safety conditions.

23.9 Software, AI and digital-twin boundaries

Predictive or AI functions may support detection and decision support, but deterministic safety logic remains authoritative. Model confidence is not equivalent to authorization. Any AI-derived recommendation entering a safety workflow should be traceable, bounded and testable against defined failure cases. Advisory computation is separated from safety authorization.

24. DESIGN REVIEW CHECKLIST

A mature review should challenge the architecture rather than merely confirm that documents exist. The following checklist is intended for G0/G1 design reviews.

Review question	Expected evidence
Does every top-level requirement have an owner?	Requirements allocation
Does every mass number have a source/status?	Mass database
Does every safety function have a verification path?	V&V; matrix
Can partial separation be detected?	State sensing design
Can a common power failure defeat multiple safety layers?	CCA
Can the capsule survive asymmetric touchdown?	Structural analysis/test plan
Is recovery deployment envelope defined?	Parachute analysis
Is AI prevented from bypassing authorization?	Kernel interface specification
Can an emergency event be reconstructed?	Immutable event record design
Are Critical Open Items (Section 20) assigned and tracked?	Open-item register + owners

25. MATURITY ROADMAP

Period / phase	Primary output	Gate
Architecture control	Requirements, interfaces, configuration baseline	G0
Analytical closure	Mass, CG, inertia, power, dynamics, safety analyses	G1
Subsystem design	Capsule, supports, separation, recovery, avionics	G2 preparation
Ground demonstration	Integrated release/recovery sequence	G3
Model validation	6-DOF correlation and uncertainty	G4
Structural/recovery tests	Crash, drop, canopy and component evidence	G5/G6
Flight development	Progressive envelope expansion	G7
Qualification/certification	Compliance evidence and controlled type design	G8

26. MASTER CONFIGURATION RECORD

Configuration item	Rev18 baseline	Status
Aircraft	HSH Heavy Safe Helicopter	LOCKED
Founder	Amirbahador Atai	CONTROLLED
Generation	Generation 1	LOCKED
Document	HSH-00	LOCKED
Revision	Rev18	BASELINE
Capsule occupancy	8 persons maximum	LOCKED LIMIT
Vehicle mass	26,000 kg	TARGET / OPEN
Capsule allocation	7,000 kg	PLANNING / OPEN
Main rotor	26.0 m	TARGET / OPEN
Tail rotor	4.0 m	TARGET / OPEN
Range	1,000 km	TARGET / OPEN
Landing supports	8 fixed energy-absorbing supports	CONCEPT / OPEN
Recovery	3 primary + 2 reserve canopies	STUDY / OPEN
Core separation	Selective, staged, non-rotor	CONCEPT / OPEN

Configuration item	Rev18 baseline	Status
Main-rotor separation	Excluded from core safety case	DEFERRED RESEARCH
Safety authorization	Deterministic Safety Kernel	CONCEPT / OPEN

27. CLOSURE PRIORITY MATRIX

Priority	Work package	Required evidence
P0	Maximum-8-person capsule closure	Bottom-up mass, geometry, CG/inertia, structural and occupant-protection evidence
P0	Stage S3 capsule release and clearance	6-DOF dynamics, clearance analysis, mechanism tests and integrated ground evidence
P0	Recovery envelope	Canopy sizing, inflation dynamics, loads, low-altitude boundary and failure-case testing
P0	Capsule crashworthiness	FEA, component tests, drop/crash testing and occupant-protection evidence
P0	Safety independence	FHA/PSSA/FMEA/CCA and fault-injection evidence
P1	26,000 kg mass closure	Component database, weighing plan, uncertainty and growth control
P1	Rotor and propulsion closure	Rotor sizing, power, OEI, transmission and thermal analyses
P1	Requirements traceability	Unique IDs, allocation, verification method and evidence register
P1	Certification basis	Applicable requirements, means of compliance and formal engagement
P1	IP due diligence	Professional prior-art, claim-scope and FTO analysis

28. CONTROLLED DOCUMENT-TREE ALIGNMENT

The Master Engineering Document Tree contains 500 sequential documents covering requirements, safety, mass properties, structures, rotor systems, propulsion, capsule, separation, avionics, performance, simulation, testing, verification, certification, IP and program management. This baseline uses that tree as the proposed maturity map rather than reproducing all 500 entries inside Document 00. Document families include 001–022, 023–044, 111–154, 157–220, 221–264, 265–330, 331–374, 399–430 and 461–500.

29. ENGINEERING INTEGRITY STATEMENT

This document remains a conceptual engineering and IP development baseline. TARGET, ASSUMPTION, STUDY, CALC and OPEN values do not become demonstrated performance merely because they appear in a detailed document. The technical framework is intended to make future verification more disciplined, not to substitute documentation for evidence. The strongest version of HSH Generation 1 is therefore not the version with the most impressive claims. It is the version in which every important claim has a visible path from requirement to model to test to controlled evidence.

30. FINAL GENERATION 1 BASELINE STATEMENT

HSH-00 Rev18 establishes the final Generation 1 conceptual master baseline for the HSH Heavy Safe Helicopter. The survival architecture centers on an occupied survival capsule, deterministic safety authorization, selective non-rotor separation, multi-mode recovery, fixed energy-absorbing landing supports and post-event survival capability. The capsule occupancy limit is fixed at 8 persons maximum for Generation 1. Future technical work shall not silently alter this value. Any proposed change requires formal configuration control and impact analysis.

The document deliberately distinguishes architecture from evidence. Mass closure, 6-DOF dynamics, separation trajectories, recovery performance, crashworthiness, safety independence, certification and patentability remain evidence-generating work packages. Critical Open Items are registered explicitly in Section 20. The appropriate next step is therefore not another expansion of Document 00. It is controlled execution of the HSH Generation 1 engineering document tree, with every major claim linked to a requirement, analysis, simulation, test result and configuration-controlled evidence record.

END OF HSH-00 REVISION 18

HSH Heavy Safe Helicopter | Generation 1 | Founder: Amirbahador Atai
 Conceptual / IP Baseline | Proprietary Concept Development Material | Not a certification approval